SUMMARY · MK 4 OPERATIONS & SUPPLY CHAIN MANAGEMENT · WEEK 1

Introduction to Operations & Supply Chain Management

What OSCM is, operations vs supply chain processes, the five supply chain processes, how services differ from goods, the goods–services continuum, the history of OSCM concepts, the triple bottom line and efficiency ratios. Opening case: Southwest Airlines and unassigned seating.

Material

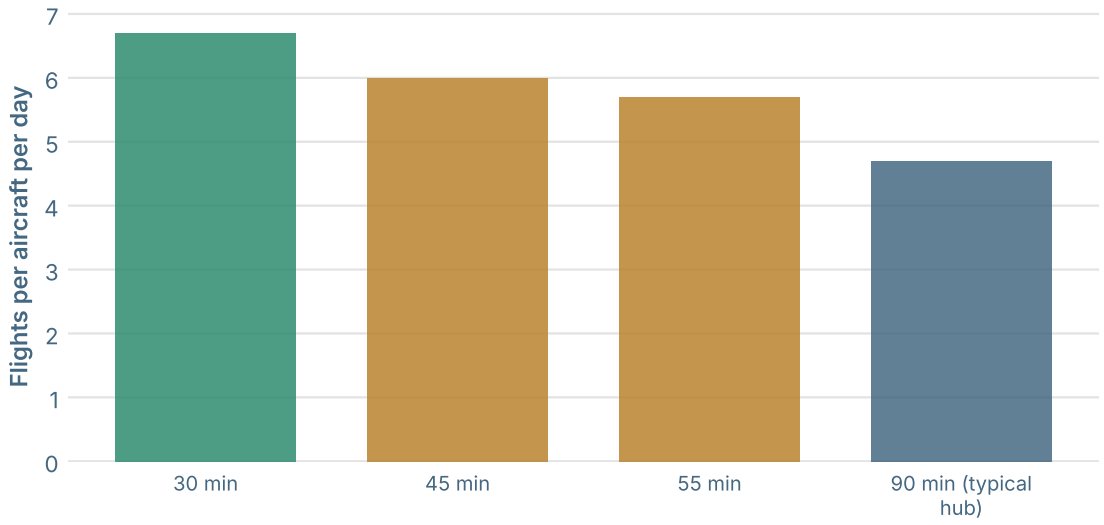
1 What is OSCM?

OSCM = design, operation and improvement of the systems that create and deliver a firm's products. A process transforms inputs into valued outputs. Operations transform resources (manufacturing or service); supply chain processes move material and information to and from them. Parka example: four pipelines, many firms — sketch the network first.

2 Southwest in numbers: ground minutes = 'free' aircraft Deep dive

Planes earn only when flying; faster turns mean more flights per aircraft.

Turn time vs flights per aircraft per day



Illustrative: 16-hour day, 1.9 h average flight.

550
Aircraft

3,400
Flights per day

~4.5 h
Ground time per day at 45-min turns

Rough math
16-hour day, 1.9 h average flight: 45-min turn → ~6.0 flights; 30-min → ~6.7; ≈350 extra flights a day across 550 jets.

3 The five supply chain processes

Plan, Source, Make, Deliver, Return — they also describe services such as hospitals (daily drug supply, scheduled procedures, staffed ER).

4 How services differ from goods

Services are intangible, require interaction, are heterogeneous, perishable, and a package of features (supporting facility, facilitating goods, explicit and implicit services). Continuum: pure goods → core goods → core services → pure

services. Product–service bundling (IBM) raises revenue but often lowers profit percentage.

5 History, sustainability and analytics

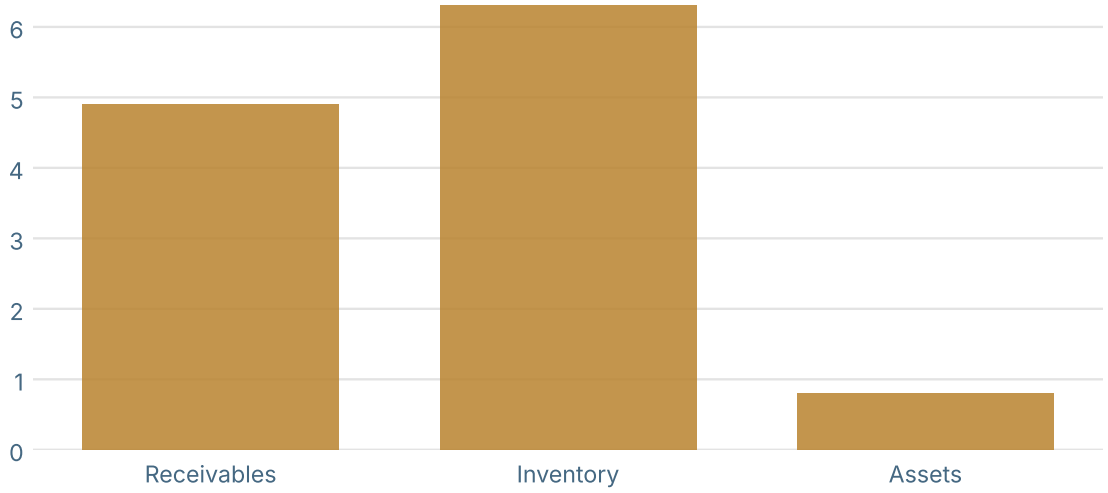
Timeline: manufacturing strategy (late 70s), JIT/lean (80s), TQM/BPR/Six Sigma (80s–90s), SCM (late 90s), e-commerce (2000s), analytics (2010s). Triple bottom line: economic, social, environmental. Analytics adds forecasting and optimisation to reporting (airline pricing).

6 Efficiency, effectiveness, value and efficiency ratios

Efficiency = lowest cost; effectiveness = right things for customer value; value = attractiveness relative to cost. Ratios: income and revenue per employee, receivables, inventory and asset turnover (most comprehensive). Honda: US$30,203 income/employee, US$585,647 revenue/employee, turnovers 4.9 / 6.3 / 0.8. GM's high asset turnover reflects bankruptcy restructuring; outsourcing inflates ratios. Always pair a ratio with an operational metric.

7 Honda's efficiency ratios charted Deep dive

Honda turnover ratios (x)



Read first

Asset turnover (0.8x).

Traps

GM's restructuring; outsourcing inflates ratios.

Key takeaways

1. OSCM = design, operation, improvement of processes.
2. Sketch the network first.
3. Plan, source, make, deliver, return.
4. Services: intangible, interactive, heterogeneous, perishable, package.
5. Efficiency ≠ effectiveness.
6. Asset turnover is the most comprehensive ratio; compare within industry.

Case studies

Case & worked answers

Southwest Airlines — why not assign seats?

550 jets, 3,400 flights/day, 30–55 minute turns. Unassigned seating speeds boarding, raising flights per aircraft (asset utilisation); passengers pay with less certainty. Below some point physical and safety limits bite. Long-haul carriers fly one leg a day and sell comfort, so the design does not transfer. Process = turnaround; metric = ground minutes; trade-off = comfort vs utilisation.

Case & worked answers

Discussion: the parka, the hospital and IBM

No single firm owns the parka's delivery date — coordination is the job; buffers decouple stages. Selling "uptime" fits when the product commoditises and the firm has service capability; measure retention and service margin. Heterogeneity and perishability make bundles hardest.

Practice questions

Try answering first. The answer key is below each question.

1. OSCM consists of...

A. Plan, do, check
B. Design, operation, improvement
C. Buy, make, sell
D. Source, make, deliver

Answer: B. As defined.

2. The process handling defective or excess goods from customers is...

A. Plan
B. Source
C. Deliver
D. Return

Answer: D. Return.

---

3. Why can't services be stored?

A. Intangible
B. Perishable — unused capacity is lost
C. Heterogeneous
D. Needs interaction

Answer: B. Empty seats vanish.

---

4. The most comprehensive efficiency ratio is...

A. Income/employee
B. Receivables turnover
C. Inventory turnover
D. Asset turnover

Answer: D. Includes all assets.

---

5. Faster turns without new planes most improve...

A. Debt ratio
B. Asset turnover
C. Current ratio
D. Payout

Answer: B. More output, same assets.

Sources

- Introduction to OSCM.pptx (MK4)
- Jacobs & Chase, Operations and Supply Chain Management, Bab 1